



Performance analysis of a novel co-generation system integrating a small modular reactor and multiple hydrogen production equipment considering peak shaving

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ABSTRACT

A novel integrated system comprising plasma gasification, small modular reactor (SMR) power generation, pressure swing adsorption (PSA) hydrogen production, and Cu–Cl cycle thermochemical hydrogen generation has been proposed for combined electricity and hydrogen co-production. In this innovative configuration, the high-temperature syngas generated by the plasma gasifier is first cooled by the steam from the SMR power plant. It subsequently undergoes water gas shift and is purified using a PSA unit to yield hydrogen product. Meanwhile, the steam generated by SMR is utilized entirely for power generation during daytime peak hours while during night time off-peak hours, a portion of the steam is diverted to provide heat for the Cu–Cl cycle, thereby facilitating load leveling. A comprehensive assessment of the hybrid system has been conducted from both thermodynamic and economic perspectives. Under two operational modes the efficiency of medical waste-to-energy efficiency reaches 63.75 % (daytime) and 63.58 % (night time), with exergy efficiencies of 45.31 % (daytime) and 44.17 % (night time). Furthermore, the net present value (NPV) of the new design is 81,727.18 k\$, dynamic payback period (DPP) of 4.34 years and an internal rate of return (IRR) of 41 %. These results substantiate the feasibility of the design from thermodynamic and economic standpoints.

Nomenclature

Symbols	
EX	exergy (MW)
P	power consumption
η	efficiency
y	year number in the project lifetime
Superscripts and subscripts	
ch	chemical
dis	discount
mw	medical waste
ph	physical
torh	plasma torches
d	day
Abbreviations	
ASU	air separator unit
DPP	dynamic payback period
HTR	high-temperature reactor
LT-WGS	low-temperature water gas shift
HX	heat exchanger
NPV	net present value
SMR	small modular reactor

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WTE	waste-to-energy
m	flow rate (kg/s)
T	temperature
ω	mass contents
des	destruction
ex	exergy
g	gasification
syn	syngas
axu	auxiliary power consumption
n	night
CHP	combined heat and power
IRR	internal rate of return
LTR	low temperature reactor
HT-WGS	high-temperature water gas shift
LHV	lower heating values
PSA	pressure swing adsorption
SGC	syngas coolers
WGS	water gas shift

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1. Introduction

In light of the escalating global concerns over environmental degradation and the urgent call to address carbon emissions [1], two critical issues emerge at the nexus: the dangerous implications of medical waste on the environment and the imperative of transitioning to cleaner energy sources [2]. The environmental hazards of medical waste have become increasingly evident, undermining conventional waste disposal methods and underscoring the need for innovative waste management approaches. The improper handling and incineration of medical waste have been correlated with the contamination of air, water, and soil and the emission of hazardous compounds that substantially threaten to ecological balance. Concurrently, pursuing effective strategies to reduce carbon emissions advances the exploration of cleaner energy alternatives in striving for a more sustainable energy paradigm [3].

Consequently, the imperative to develop efficient, safe, and sustainable waste treatment technologies stands as a critical concern within the scientific and industrial realms. Plasma gasification technology has captured significant interest recently and has been touted for its distinctive capacity to manage hazardous solid waste [4]. As a process characterized by high temperatures and intense energy input, plasma gasification converts waste material into a plasma state, disintegrating both organic and inorganic substances to their fundamental elements [5]. This method is beneficial in diminishing the volume of hazardous solid waste and decomposing harmful compounds, thus reducing the risk of environmental pollution. Additionally, conventional waste incineration power plants demonstrate relatively lower electrical efficiency, with figures ranging from 14 % to 28 % [6]. Nevertheless, achieving the high-efficiency profile of plasma gasification necessitates significant investment and operational costs. These expenses could be compensated for by the synergistic integration with SMR, which further enhances the overall efficiency of the process.

In contrast to traditional nuclear reactors, SMRs feature simplified, standardized designs and inherent safety characteristics [7]. Nevertheless, SMRs typically generate steam at substantially lower temperatures than coal-fired plants resulting in comparatively lower thermoelectric conversion efficiency [8]. Integrating SMRs with plasma gasification improves energy efficiency through the utilization of heat from synthesis gas, which raises the temperature of steam and feedwater in SMR power plants. This higher operating temperature enables the turbines in SMR power plants to generate more electricity.

Modular nuclear power units are primarily designed to provide a consistent supply of base-load electricity, with an emphasis on optimized design and operation tailored to meet steady power demands. However, this inherent stability limits their capacity to rapidly modulate power outputs in response to sharp fluctuations in electricity demand [9]. The limited flexibility of modular nuclear units in the face of significant demand swings could lead to grid instability. It is important to note that the reduced load-following capacity of nuclear reactors is not solely a result of thermal inertia; rather, it is intimately connected to the reactor's predetermined operational parameters. These parameters are orchestrated for optimal performance under specific conditions, and adjustments to reactor inlet and outlet conditions to alter these parameters can lead to performance inefficiencies. While power output can be modulated by decreasing operational intensity, such adjustments typically do not translate to proportional reductions in production costs or extend the fuel lifecycle. Therefore, from both economic and technological perspectives, modifying reactors to enhance load-following is not generally advised [10]. If modular nuclear power primarily focuses on stable base-load electricity generation, investments to improve load-following capability may not provide equal returns. The aforementioned statement elucidates the intrinsic complexities involved in the integration of load-following capabilities into modular nuclear reactors. It underscores the imperative for comprehensive assessments that span technical aptitude, safety protocols, and financial feasibility.

Such assessments are crucial for assessing the feasibility and viability of integrating load-following functions into modular nuclear power systems. The synergistic integration with the Cu–Cl cycle for hydrogen production offers a promising solution to mitigate load-following challenges in modular nuclear power, enabling efficient load management.

An increasing number of scholars have dedicated their efforts to the research of waste-to-hydrogen conversion and nuclear-based hydrogen production. Chien et al. [11] dynamically evaluated Pakistan's green hydrogen potential from wind and solar via Fuzzy Hierarchical Analysis and Data Envelopment Analysis, favoring wind as a viable, sustainable energy solution. Wijayasekera et al. [12] assessed factors impacting waste-to-hydrogen production, analyzing their variability and effects on economics, the environment, and facility development. Minutillo et al. [13] analyzed the efficiency of plasma gasification technology in treating solid waste and converting it to energy through thermochemical modeling, and also evaluated the performance, which was shown to be superior to conventional waste incineration technologies. Wang et al. [14] introduce a solar-nuclear power system using a solar tower and a modular reactor, enhancing output with a supercritical CO₂ Brayton cycle, proven stable by simulation. Yin et al. [15] combined plasma gasification with ionic CO₂ capture for hydrogen from medical waste, examining factors affecting syngas and efficiency. Martins et al. [16] evaluated three gasification methods for green hydrogen; supercritical water gasification had the highest yield yet was economically unfeasible, requiring cost-reducing optimizations. A comparison of the current study with the existing literature is displayed in Table 1.

The current research landscape lacks a comprehensive energy production system that is modular, flexibly deployable, and environmentally friendly, advancing both the resource recovery and harmless utilization of hazardous waste while reducing reliance on fossil fuels. This study adopts static planning and innovatively proposes an integrated system, a novel fusion of plasma gasification, SMR, PSA for hydrogen production, and the Cu–Cl cycle. SMR is ideal for optimal heat integration with a plasma gasifier due to its low central steam temperature. The syngas produced by plasma gasification is used for hydrogen production, and its heat is also efficiently utilized by the steam from the modular nuclear power plant, which achieves a two-way value-added energy. During peak periods, all the steam is used to generate electricity to meet the demand of the grid, while during low periods, some of the steam is diverted to supply the heat demand of the hydrogen production process in the Cu–Cl cycle. The modular aspect of plasma gasifiers and SMRs offers significant advantages in scalability, flexibility, and ease of integration. The modular design simplifies maintenance, facilitates transportation and installation, improves significant cost-effectiveness, and reduces deployment time in different geographical locations. The proposed integrated system not only takes advantage of the inherent

Table 1
Comparison of the proposed study.

Reference	Process	Comparison with current study
[11]	Dynamic planning for different renewable energy sources	Fast calculations, simplified decision-making process, easy to understand, but lack of adaptability, over-reliance on assumptions.
[12]	multiple waste-to-hydrogen approaches	Considering the energy analysis and exergy analysis of the process, it is possible to identify the parts that need to be optimized.
[13]	Integrated plasma gasification combined cycle	The modular design of the system provides flexibility and facilitates construction and maintenance.
[14]	Solar energy + SMR	Power load balancing with grid connection is considered
[15]	Medical waste plasma gasification to hydrogen	This paper considers sensitivity analysis under the influence of key parameters
[16]	Plasma gasification to hydrogen	The system in this study has a positive NPV and is more economically viable relative to those in reference 19.

carbon emission-free benefits of SMR but also achieves resourceful and harmless treatment of hazardous waste through plasma gasification, which significantly enhances its environmental friendliness, reduces dependence on fossil energy and provides a more sustainable method of energy production, deliberately designed to facilitate the transition towards low-carbon energy sources. The proposed system is a new type of system that combines multiple functions such as environmental friendliness, flexibility, and maintenance of grid stability.

2. System description

A hybrid system integrating plasma gasification and modular nuclear power as well as multiple hydrogen production systems has been

proposed, which is somewhat capable of achieving harmless treatment of medical waste as well as assisting efficient peaking of modular nuclear power, as sketched in Fig. 1. The suggested hybrid system comprises four distinct subsystems, which are plasma gasification, SMR power plant, Cu–Cl cycle and Pressure swing adsorption (PSA).

The plasma gasification system is the first part of the hybrid system proposed in this paper. The plasma gasifier can efficiently and harmlessly convert medical waste into high-quality syngas, then feed as feedstock to the pressure swing adsorption subsystem. At the base of the plasma gasifier, two thermal plasma torches are installed to produce the required thermal plasma, providing the essential heat for waste molecule decomposition within the gasifier furnace. The temperature inside the furnace is maintained between 2500 and 4500 °C [17]. The

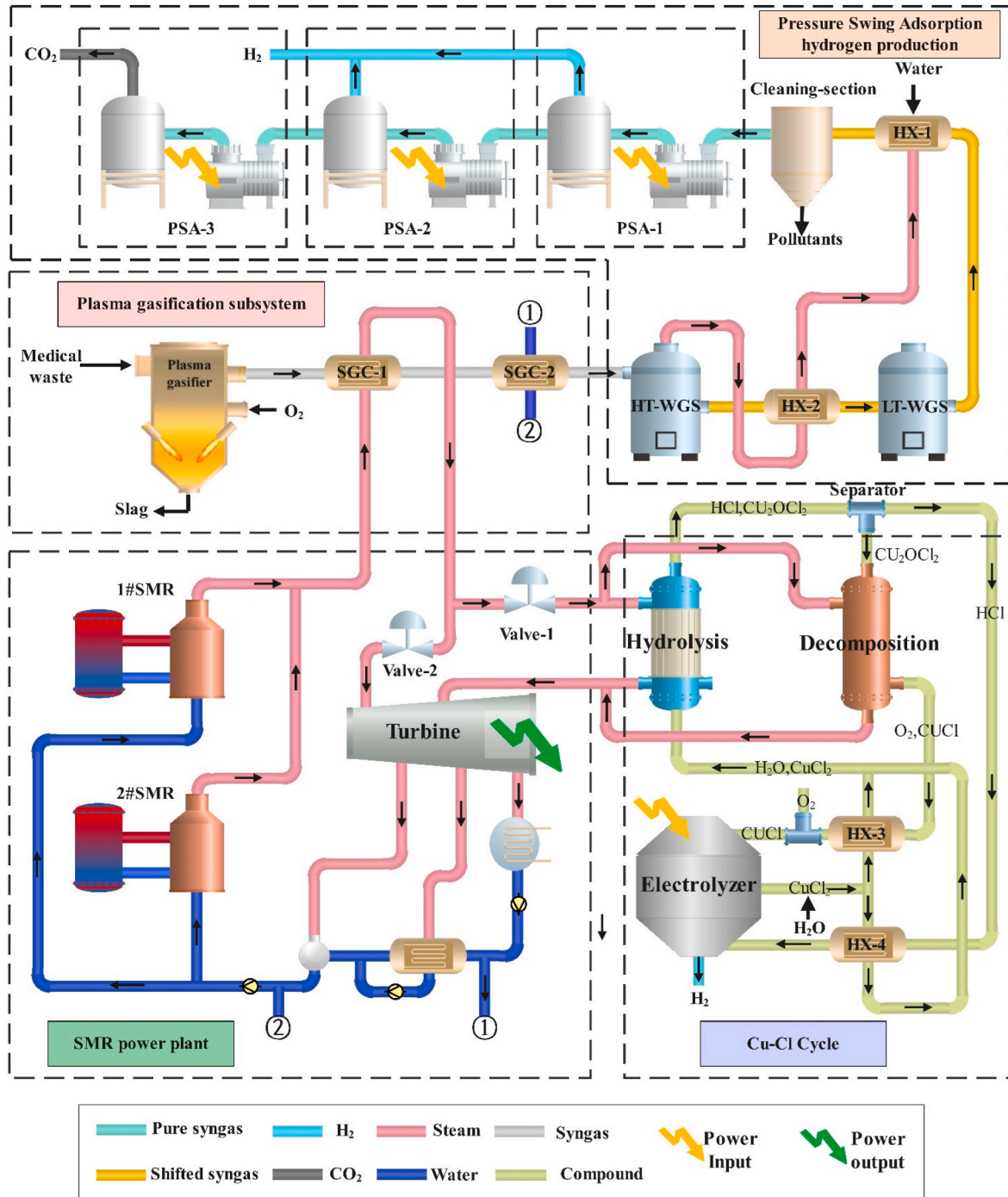


Fig. 1. Diagram of the proposed system.

generation of harmful substances such as dioxins and furans during the gasification process can be suppressed by the high temperature inside the furnace. Simultaneously, the organic matter in the feed is decomposed into useable fuel [18]. The inorganic material in the feed is melted into a glassy form that can be used as a building material [19]. In the hybrid system proposed by this study, the plasma gasification subsystem serves as a crucial initial component. This subsystem transforms medical waste—particularly types resistant to conventional treatment—into high-caliber syngas, which subsequently supplies to the pressure swing adsorption subsystem for hydrogen production.

The raw syngas generated by the plasma gasifier exhibits a substantially elevated temperature, peaking at 1072 °C. Before the syngas can be utilized, it needs to be cooled down. Reducing syngas temperature to 400 °C is imperative to accommodate the water-gas shift reaction. Notably, the thermal energy of syngas is reclaimed to bolster the overall efficiency of the system. Within the framework of the SMR power plant, this thermal energy is channelized to preheat the primary steam and feedwater, thus capitalizing on the heat generated.

Subsequent to cooling, the syngas is amalgamated with steam from heat exchanger 2 (HX-2) and channeled into the two-stage water gas shift (WGS) reactor, catalyzing its conversion into a hydrogen-rich gas. Owing to the exothermic characteristics of the WGS reaction, HX-2 and heat exchanger 1 (HX-1) have been positioned downstream of the high-temperature water gas shift (HT-WGS) reactor and the low-temperature water gas shift (LT-WGS) reactor, respectively. This arrangement aims to effectively capture and utilize the heat generated during the WGS reaction [20], thereby heating water to engender steam requisite for the process, and ensuring efficient energy management. Subsequently, in order to protect the PSA unit and to improve plant efficiency, undesirable constituents within the modified syngas, including COS, NH₃, H₂S, and moisture, are excised in the cleaning unit. The resultant purified syngas is then conveyed to the tertiary PSA unit to produce the hydrogen and carbon dioxide via pressure swing adsorption.

The novel hybrid system features bifurcated operational modes, tailored to the diurnal fluctuations in electricity demand: the daytime operation mode during the peak electricity consumption period and the nighttime operation mode during the low electricity consumption period. The crux of differentiation between these two operational states resides within the operational dynamics of the SMR power plant segment.

(a) Operating mode of the system at daytime

During the daytime hours of peak electricity consumption, Valve-1 is completely shut down and all of the steam generated from the SMR is used to generate electricity to power the grid.

(b) Operating mode of the system at night

During nocturnal phases characterized by diminished electricity demand, Valve-1 is actuated, while Valve-2 is adjusted to a partially closed state. A proportion of steam from the SMR unit is diverted to supply thermal energy for the hydrolytic and thermal decomposition reactions within the Cu–Cl cycle. This process effectuates a reduction in electric output from the SMR facility while facilitating peak load adaptation. After providing heat to the Cu–Cl cycle, the steam, retaining a temperature of 287 °C, is subsequently routed back to the steam turbine for power generation. The Cu–Cl cycle is sustained by the introduction of water, which undergoes hydrolysis, thermal decomposition, and electrolysis, mediated by Cu and Cl ions, to bifurcate into hydrogen and oxygen. This scheme effectively transmutes a fraction of the electrical output from the SMR installation into hydrogen, thus enabling a peak load-shifting capability within modular nuclear systems.

With the above new hybrid system, modular nuclear power can achieve efficient peak shifting and, at the same time, the conversion of hazardous medical waste to clean and safe energy.

3. Methodology

3.1. Model development

The proposed novel hybrid system is co-simulated by Aspen plus and EBSILON Professional as sketched in Fig. 2. The plasma gasification subsystem, the PSA hydrogen production subsystem, and the Cu–Cl cycle subsystem were modeled by Aspen plus [21]. EBSILON Professional was employed to model the SMR power plant [22].

3.1.1. Plasma gasification subsystem

Plasma gasification offers a promising method for medical waste disposal, employing high temperatures for pyrolysis and electron detachment to create a plasma state. The process, simulated by the DECOMP chemical stoichiometry and high-temperature reactor (HTR) modules, breaks down waste into basic elements such as C, H, O, N, S, H₂O, and ash. These elements are quantified using a state calculator in Fortran. The HTR module, operating at roughly 2500 °C, provides the necessary thermal energy, driving reactions like dissociation, oxidation, and reduction within the plasma. This leads to the production of gaseous byproducts such as hydrogen, carbon monoxide, methane, forming syngas, with key reactions detailed in Table A1 of the appendix.

3.1.2. PSA subsystem

The produced syngas is fed to the WGS reactor for further reactions, containing a catalyst based on cobalt-molybdenum (CO/MO). The reactions that occur within the WGS reactors are shown in Equation (A13) and Equation (A14) in the Appendix.

The H₂-rich gas produced by the reaction is cooled to 35 °C and compressed to 30 atm using a multi-stage compressor. The H₂-rich gas is then sequentially processed through PAS-1, PSA-2, and PSA-3 units to separate hydrogen gas from carbon dioxide, with the carbon dioxide being sequestered. The operating parameters for the PSA subsystem are detailed in Table 2.

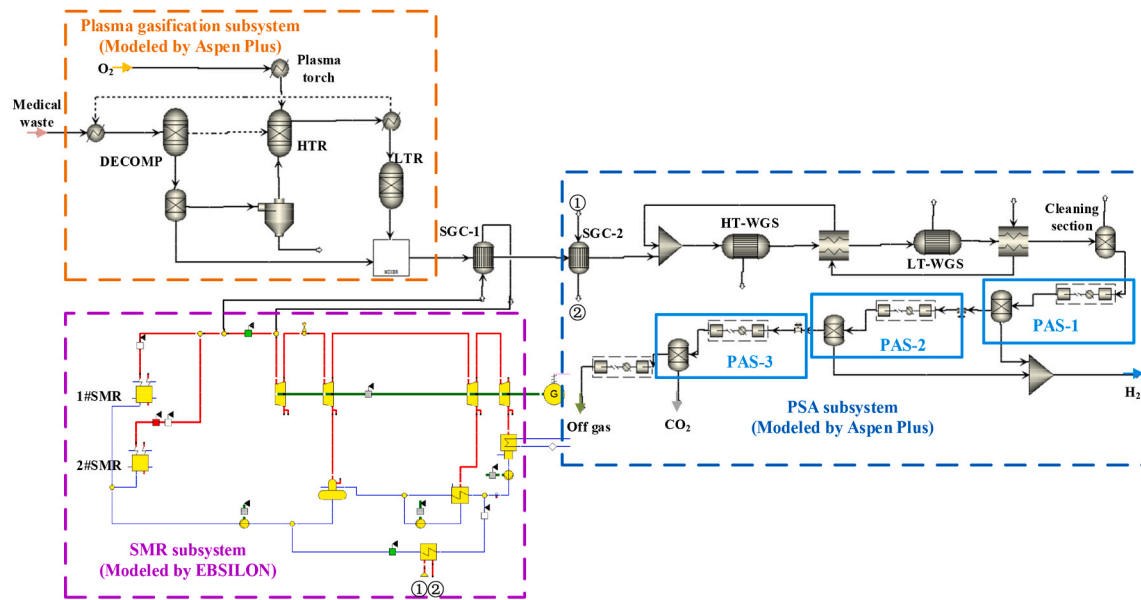
3.1.3. Cu–Cl cycle subsystem

The hydrogen production methodology employed in this study is the three-step Cu–Cl cycle, comprising distinct hydrolysis, decomposition, and electrolysis reactions. Each reaction stage is conducted within dedicated chemical stoichiometric reactors under specific conditions. The reactions and their corresponding specified conditions are outlined as follows. The operating parameters of the Cu–Cl cycle are shown in Table 3.

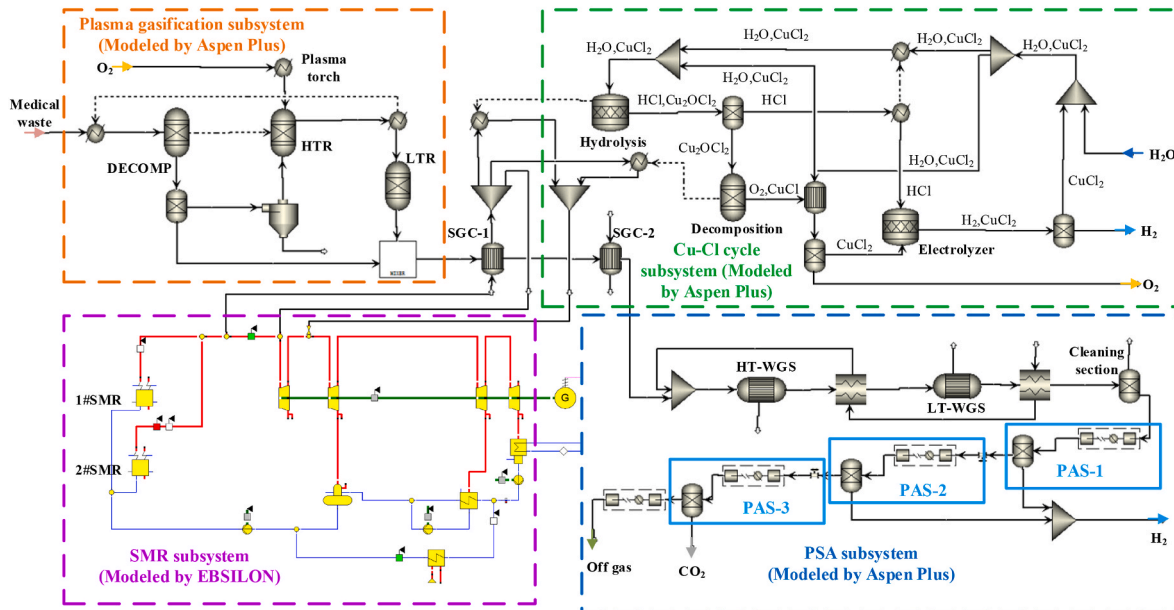
The chemical equation (A15) in Appendix presented represents the hydrolysis reaction, which is simulated by the “Hydrolysis” chemical stoichiometric reactor module. The hydrolysis reaction is conducted within an essential thermal window of 350–400 °C, and it proceeds under ambient pressure conditions. The required thermal energy for this reaction is provisioned by steam generated from the SMR power plant [23].

The decomposition reaction, characterized by the chemical equation (A16) in Appendix, is simulated through employment of the chemical stoichiometric reactor module termed “Decomposition”. This reaction occurs typically within the temperature range of 500–540 °C and operates under ambient pressure. The reactant for the decomposition reaction, Cu₂OCl₂ generated from the hydrolysis reaction, is introduced into the reactor for the designated reaction sequence.

Following the cooling of the mixture derived from the hydrolysis reaction, comprising HCl and H₂O, as well as the Cu–Cl produced from the decomposition reaction, the resulting mixture is introduced into the chemical stoichiometric reactor named “Electrolysis” to initiate the electrochemical reaction. This reaction yields H₂ as the ultimate product. The electrolysis reaction occurs under standard temperature conditions and at a pressure of 25 bar. The equations for the above three reactions are shown in Equations A15–16 in the Appendix.



(a) Operating mode of the system at daytime



(b) Operating mode of the system at night

Fig. 2. Simulation models of the proposed hybrid system.

Table 2

Operating parameters of the PSA subsystem [20].

Item		Value
HT-WGS	Temperature (°C)	400
	Pressure (MPa)	0.10
LT-WGS	Temperature (°C)	200
	Pressure (MPa)	0.1
Pressure swing adsorption (PSA)	Temperature (°C)	35
	Pressure (MPa)	3.04
PSA-1	H ₂ recovery	95 %
PSA-2	H ₂ recovery	95 %
PSA-3	H ₂ and CO ₂ recovery	0.5 %/90 %

Table 3

Operating parameters of the Cu-Cl cycle [23].

Item		Value
Hydrolysis	Temperature (°C)	400
	Pressure (MPa)	0.1
Decomposition	Temperature (°C)	500
	Pressure (MPa)	0.1
Electrolysis	Temperature (°C)	25
	Pressure (MPa)	2.5

3.2. Simulation and validation of models

Within the scope of this research, the corroboration of models pertaining to plasma gasification and SMR power plant modules is carried out, thereby enabling emulation of the interactive behavior of these systems under a variety of operating conditions. Through a comparison of the simulated outcomes with design parameters documented in the reference literature, a congruence between the simulated results and benchmark values is evident, as illustrated in Table 4. This validation procedure enhances the dependability of research outcomes and lends credence to the reliability of the study's findings.

3.3. Basic assumptions

Prior to conducting thermal and economic analyses in this paper, several fundamental assumptions are proposed to ascertain the advantages of the novel hybrid system in comparison with the reference SMR power plant. These assumptions serve the dual purpose of simplifying modeling and analysis procedures.

- In the thermal-chemical reactions occurring within the system, reactants are assumed to be thoroughly mixed.
- The chemical reactions are considered to be complete.
- The thermal power generated by the SMR process and the efficiency of the SMR power plant stay consistent.
- The primary steam parameters of the SMR process are held unchanged.
- The environmental temperature and pressure are constantly 25.0 °C and 101.325 kPa.
- Kinetic and potential energy changes are ignored [26].
- Without consideration of the kinetic and potential exergy. Consequently, the medical waste-to-energy process can be independently examined, while the performance assumptions of the SMR power generation process are maintained unchanged.

3.4. Energetic evaluation

For effective evaluation of the performance of the medical waste plasma gasification process, the plasma gasification efficiency, denoted

Table 4
Validation results for plasma gasification and SMR power plant models.

Item			Reference	Simulation
SMR power plant (modeled by EBSILON Professional, validated by the data in Reference [24])	Main steam (into turbine)	Temperature (°C)	440.00	440.00
		Pressure (MPa)	4.00	4.00
		Flow rate (kg/s)	3.47	3.47
	Feedwater (into steam generator)	Temperature (°C)	104.00	103.56
		Pressure (MPa)	5.00	5.00
		Flow rate (kg/s)	3.47	3.47
		Electric power (MW)	3.4	3.38
		Electric efficiency (%)	34.00	33.80
	Composition of syngas (vol%)	H ₂	43.50	43.53
		CO	34.50	34.01
Plasma gasifier (modeled by Aspen Plus, validated by the data in Reference [25])		CH ₄	0.01	0.16
		CO ₂	0.03	0.35
		N ₂	5.63	5.65
		H ₂ O	16.22	16.20
		H ₂ S	0.09	0.09
		COS	0.01	0.01
		Sum	100	100
	Temperature of syngas (°C)		1267.00	1254.50
	Lower heating value (LHV) of syngas (MJ/kg)		13.44	13.21

by η_g , is delineated. η_g is determined by the quotient of the chemical power of the syngas and the chemical power of all direct and indirect input fuels. This efficiency quantifies the energy conversion from medical waste into the energy content of the produced syngas. The ratio between the waste feed and the gasification agent is the critical parameter of the plasma gasification process. Reducing this parameter augments the exothermic reaction magnitude, concomitantly diminishing the power necessity for the plasma torch and reducing the LHV of the syngas output. Concurrently, electricity is consumed by both the plasma torch and the Air Separation Unit (ASU), providing the necessary thermal energy for the plasma gasifier and generating O₂ for use as the gasification agent. The plasma gasification efficiency is computed according to the following expression. The introduction of the plasma gasification efficiency metric facilitates a comprehensive evaluation of the conversion process and its energy utilization aspects, which is crucial for enhancing the sustainability and effectiveness of medical waste treatment [27].

$$\eta_g = \frac{m_{\text{syn}} \times \text{LHV}_{\text{syn}}}{m_{\text{mw}} \times \text{LHV}_{\text{mw}} + \frac{P_{\text{tor}} + P_{\text{ASU}}}{\eta_{\text{tor}} \times \eta_{e,n}}} \quad (1)$$

where m_{syn} and m_{mw} are the crude syngas flow rate and medical waste flow rate, kg/s; LHV_{syn} and LHV_{mw} are the lower heating values of the syngas and medical waste, kJ/kg; P_{tor} is the total power consumption of the plasma torches, kW; P_{ASU} is the power consumption of the air separator unit; η_{tor} is the plasma torch efficiency, which is chosen as 90 % [28]; $\eta_{e,n}$ is the net electrical efficiency of the SMR power plant which can be defined as the following equation.

$$\eta_{e,n} = \frac{P_{\text{SMR}}}{Q_{\text{SMR}}} \quad (2)$$

Where P_{SMR} is the net power of SMR power plant, kW; Q_{SMR} is the input heat of SMR, kW.

The auxiliary power consumption of the novel hybrid system comprises the power consumption of the ASU P_{ASU} , plasma torch power consumption P_{torch} , power consumption of compressors at various stages within the Pressure Swing Adsorption (PSA) subsystem P_{PSA} , power consumption of the Cu–Cl electrolysis process $P_{\text{Cu–Cl}}$. The auxiliary power consumption of the novel hybrid system during daytime $P_{\text{aux,d}}$ is computed using Formula 2, while the auxiliary power consumption during nighttime $P_{\text{aux,n}}$ is calculated using Formula 3. This delineation underscores the various components contributing to auxiliary power consumption, offering a comprehensive perspective on the energy dynamics of the innovative hybrid system.

$$P_{\text{aux,d}} = P_{\text{ASU}} + P_{\text{torch}} + P_{\text{PSA}} \quad (3)$$

$$P_{\text{aux,n}} = P_{\text{ASU}} + P_{\text{torch}} + P_{\text{PSA}} + P_{\text{Cu–Cl}} \quad (4)$$

In the context of the retrofitting project for an SMR power plant, where the power generation capacity of the SMR remains unchanged, the electrical power generated from medical waste can be determined by subtracting the SMR power generation capacity from the overall net electrical power generation. During the daytime operational mode, medical waste is transformed into synthetic gas through a series of reactions, leading to the generation of H₂. Thus, the energy derived from the conversion of medical waste can be quantified using the following equation:

Operating mode of the system at daytime

$$P_{\text{mw,d}} = P_{\text{tot}} - P_{\text{SMR}} + m_{\text{H}_2, \text{PSA}} \times \text{LHV}_{\text{H}_2} - P_{\text{aux,d}} \quad (5)$$

Where $P_{\text{mw,d}}$ is the energy derived from the conversion of medical waste during the daytime operational mode, kW; P_{tot} is the total net power of the proposed system, kW; $m_{\text{H}_2, \text{PSA}}$ is the flow rate of H₂ produced from the PSA unit, kg/s; LHV_{H_2} is the LHV of H₂, kJ/kg.

Due to the nighttime operational mode, a portion of the steam is

utilized to provide heat for the Cu–Cl cycle. As a result, the energy derived from the conversion of medical waste can be determined using the following formulation:

Operating mode of the system at night

$$P_{mw,n} = P_{tot} + m_{H_2,Cu-Cl} \times LHV_{H_2} - P_{SMR} + m_{H_2, PSA} \times LHV_{H_2} - P_{aux,n} \quad (6)$$

where $P_{mw,n}$ is the energy derived from the conversion of medical waste during the night operational mode, kW; $m_{H_2,Cu-Cl}$ is the flow rate of H_2 produced from the Cu–Cl cycle, kg/s.

In order to assess the overall performance of the novel hybrid system, the efficiency of medical waste-to-energy conversion is defined as follows:

$$\eta_{mw,d} = \frac{P_{mw,d}}{m_{mw} \times LHV_{mw}} \quad (7)$$

$$\eta_{mw,n} = \frac{P_{mw,n}}{m_{mw} \times LHV_{mw}} \quad (8)$$

where $\eta_{mw,d}$ is the medical waste-to-energy efficiency in daytime operating mode, %; $\eta_{mw,n}$ is the medical waste-to-energy efficiency in night operating mode, %; m_{mw} is the flow rate of medical waste, kg/s; LHV_{mw} is the LHV of medical waste, kJ/kg.

3.5. Exergy evaluation criteria

Exergy analysis provides a more precise assessment of the effective utilization of energy within a system, extending beyond mere energy conversion efficiencies. This aids in identifying the specific origins of energy losses and wastage [29]. Moreover, Exergy analysis unveils inherent system imperfections, dormant energy, and avoidable losses, thus guiding improvement measures aimed at enhancing the efficiency of the system.

The exergy of medical waste can be computed using the formula commonly applied for solid fuel exergy calculations, as presented in the following equation [30].

$$EX_{mw} = m_{mw} \times LHV_{mw} \times \left(1.0064 + 0.1519 \times \frac{\omega_H}{\omega_C} + 0.0616 \times \frac{\omega_O}{\omega_C} + 0.0429 \times \frac{\omega_N}{\omega_C} \right) \quad (9)$$

where EX_{mw} is the exergy of the medical waste, MW; m_{mw} is the flow rate of the medical waste, kg/s; ω_C , ω_H , ω_O , ω_N are the mass contents of carbon, hydrogen, oxygen, and nitrogen in the medical waste, %.

The exergy of the syngas generated from medical waste through the plasma gasification process can be calculated using the following equation.

$$EX_{syn} = EX_{syn}^{ph} + EX_{syn}^{ch} \quad (10)$$

where EX_{syn}^{ch} and EX_{syn}^{ph} are the chemical exergy and physical exergy of the syngas, MW.

The exergy balances of each component, subsystem, and the entire system are computed using Equation (11), while their efficiencies are calculated using Equation (12).

$$\sum EX_{in} + \sum W_{in} = \sum EX_{out} + \sum W_{out} + \sum EX_{des} \quad (11)$$

$$\eta_{ex} = 1 - \frac{\sum EX_{des}}{\sum EX_{in}} \quad (12)$$

where $\sum EX_{in}$ and $\sum EX_{out}$ are the total exergy input and total exergy output, kW; $\sum W_{in}$ and $\sum W_{out}$ are the total work input and total work output, kW; $\sum EX_{des}$ is the total exergy destruction, kW.

3.6. Economic evaluation criteria

Economic analysis is a pivotal tool for informed decision-making, optimal resource utilization, and project feasibility assessment, contributing to the sustainability and long-term success. The NPV method is a typical capital budgeting technique that assesses investment project feasibility by discounting future cash flows to their present value, considering the time value impact for comparing cash flows from different periods. It includes all investment-related cash flows, covering initial expenses and future returns, enabling a thorough economic viability assessment of investments [31].

$$NPV = \sum_{y=1}^k \frac{C_{in} - C_{out}}{(1 + r_{dis})^y} \quad (13)$$

where y is the year number in the project lifetime; r_{dis} is the discount rate; C_{in} and C_{out} are the cash inflow and cash outflow in year y , k\$; k is the project lifetime, year.

DPP is a capital budgeting method assessing project recovery time while considering the time value of money. It discounts future cash flows for a precise fund value. It offers a more comprehensive analysis than traditional payback periods by factoring in future cash flows. Additionally, DPP's calculation includes cash flow distribution, aiding in risk assessment by showing variations in payback periods under different conditions [31].

$$\sum_{y=1}^{DPP} \frac{C_{in} - C_{out}}{(1 + r_{dis}\%)^y} = 0 \quad (14)$$

The IRR is a financial metric assessing investment profitability, representing the annualized compound rate of return for projects with varying cash flows. Based on the present value of the cash flows and the real rate of return target, the discount rate is continuously adjusted until a discount rate is found which makes the sum of the present values of the cash flows equal to zero. When the sum of the present values of the cash flows is equal to zero, this discount rate is the desired internal rate of return, IRR. IRR helps determine project worthiness; if it exceeds the investor's required rate of return, the project is considered attractive as it yields returns above expectations [32].

$$0 = \sum_{y=1}^k \frac{C_{in} - C_{out}}{(1 + IRR)^k} \quad (15)$$

4. Results and discussion

4.1. Parameters of proposed system

In this study, medical waste from Ref. [33] was chosen as the feed-stock for plasma gasification. The composition of the medical waste is presented in Table 5, and both industrial analysis and elemental analysis were conducted. The moisture content of the medical waste was measured at 7.04 %, attributed to the presence of high-water content components such as visceral organs within the medical waste. The LHV

Table 5
Properties of the medical waste (as received basis) [33].

Item	Value
Proximate analysis (wt.%)	Moisture
	82.37
	Fixed carbon
	8.70
Ultimate analysis (wt.%)	Ash
	1.89
	C
	44.64
	H
	6.56
	O
LHV (MJ/kg)	N
	0.2
	S
	0.07
	15.57

of the medical waste was determined as 15.57 MJ/kg. The incoming rate of medical waste was 0.83 kg/s. O₂ was selected as the gasifying agent and introduced into the plasma gasification reactor as a high-temperature plasma jet. Within the reactor, medical waste underwent conversion into raw syngas, yielding a production rate of 1.10 kg/s. The LHV of the generated crude syngas was calculated as 10.92 MJ/kg. The parameters of the plasma gasifier, along with the characteristics of the produced crude syngas, are detailed in Table 6 and Table 7, respectively.

The raw syngas, upon undergoing cooling, proceeds into a two-stage water-gas shift reaction, yielding H₂-rich gas. To mitigate the potential hazards of certain gases within the syngas to the PSA hydrogen production equipment and to enhance operational efficiency, the H₂-rich gas undergoes purification through a cleaning unit before entering the PSA unit. This step aims to remove undesired constituents. Subsequently, the processed gas enters a third-stage PSA unit to generate product H₂ and CO₂. An H₂ flow rate of 0.1 kg/s and a CO₂ flow rate of 1.2 kg/s were generated during the process. The parameters associated with the produced H₂ and CO₂ are presented in Table 8.

During the nocturnal operational mode, a portion of steam from the SMR power plant is allocated for the endothermic water splitting and decomposition reactions within the Cu–Cl cycle. A flow rate of 0.14 kg/s of H₂O is introduced into the Cu–Cl cycle, undergoing a series of three sequential reactions, producing 0.016 kg/s of H₂ and 0.125 kg/s of O₂ as final products. The parameters of the Cu–Cl cycle are shown in Table 9.

The SMR system generates steam at a temperature of 440 °C, with a cumulative steam flow rate of 6.94 kg/s across two SMR units. Subsequent passage through the syngas cooler 1 (SGC-1) increases the temperature, elevating the primary steam temperature to 523 °C. Under the daytime operational mode, the plant attains an electrical power output of 8 MW. Conversely, the nighttime mode employs a distinctive operational scheme wherein half of the steam, accounting for 3.47 kg/s, is diverted to facilitate the Cu–Cl cycle for auxiliary heating before being reintegrated into the power generation process. Consequently, the nocturnal power generation capacity is observed to reach 6.03 MW. This study undertakes a comprehensive comparative assessment of pivotal performance parameters across two distinct operational modes, as delineated in Table 10.

4.2. Energy analysis

Energy analysis constitutes a pivotal approach to comprehending the performance of systems. Dual operational modes of the proposed hybrid system are subjected to energy analysis, yielding outcomes as presented in Table 11. Under the daytime operational mode, the integration of the plasma gasifier with the SMR plant results in an augmented electrical output owing to the heating of SMR primary steam by syngas. However, the incorporation of components like the plasma gasifier and PSA unit introduces supplementary energy consumption to the entire system. Furthermore, the energy-intensive nature of the plasma torch contributes to a relatively substantial increase in auxiliary energy consumption compared to the conventional SMR power plant. Notably, within this

Table 6
Parameters of the plasma gasifier.

Item	Value
Medical waste feed rate (kg/s)	0.80
Inlet oxygen	Flow rate (kg/s) Temperature (°C)
	0.32 25.00
Outlet syngas	Flow rate (kg/s) Temperature (°C)
	1.10 1070.73
Plasma torch	Efficiency (%) Power consumption (MW)
	90.00 1.62
ASU consumption	0.31
Plasma gasification efficiency (%)	63.41

Table 7
Parameters of the crude syngas.

Item	Value
Crude syngas	Temperature (°C) Pressure (MPa) Flow rate (kg/s)
	1072.31 0.10 1.10
Composition of crude syngas (wt.%)	H ₂ N ₂ CO H ₂ O CO ₂ H ₂ S
	3.75 0.14 63.46 13.94 18.64 0.07
LHV (MJ/kg)	10.92

Table 8
Parameters of the production.

Item	Value
H ₂	Temperature (°C) Flow rate (kg/s)
	35 0.10
CO ₂	Temperature (°C) Flow rate (kg/s)
	35 1.2

Table 9
Parameters of the Cu–Cl cycle.

Item	Value
Inlet water	Temperature (°C) Flow rate (kg/s)
	25 0.14
H ₂ production	Temperature (°C) Flow rate (kg/s)
	25 0.016
O ₂ production	Temperature (°C) Flow rate (kg/s)
	25 0.125

Table 10
Parameters of the SMR power plant.

Item	Value	
	day	night
Temperature of primary steam (°C)	440.00	440.00
Pressure of primary steam (MPa)	4.00	4.00
Flow rate of primary steam (kg/s)	6.94	6.94
Inlet temperature of the steam turbine (°C)	523.00	523.00
Inlet pressure of the steam turbine (MPa)	3.995	3.995
Electricity production (MW)	8.00	6.03
Steam extraction to Cu–Cl cycle (kg/s)	0	3.47

Table 11
Energy performance of the proposed hybrid system.

Item	Reference	Proposed hybrid system	
		day	night
Gross power of SMR power plant (MW)	7.17	8.00	6.03
Total auxiliary power consumption (MW)	0.37	4.33	4.54
H ₂ production (MW)	/	11.02	12.93
Waste to energy (MW)	/	7.94	7.92
Plasma gasification efficiency η_g (%)	/	63.41	63.41
Waste-to-energy efficiency (%)	/	63.75	63.58
Total process efficiency (%)	34.00	45.45	44.57

hybrid system, medical waste conversion yields an energy of 7.94 MW, achieving a conversion efficiency of 63.75 %, surpassing similar hybrid systems. The plasma gasification efficiency herein stands at 63.41 %, reflecting exemplary performance. The newly designed total process efficiency reaches 45.45 %, significantly outperforming the conventional SMR power plant. Conversely, during the nighttime operational mode, a portion of steam is allocated for Cu–Cl cycle heating, leading to

a decrease in SMR power plant electricity generation. However, the clean energy production of H_2 experiences an upsurge, contributing to grid load balancing. Notably, the nighttime operational mode exhibits a medical waste-to-energy efficiency of 63.58 %, with the hybrid system's total process efficiency reaching 44.57 %. Both operational modes show minimal variations in efficiency parameters, demonstrating that the hybrid system contributes to grid load modulation without significant efficiency losses due to load balancing. This underscores the advanced design and performance of the hybrid system, making it suitable for efficient grid load balancing and safe medical waste treatment. This innovative design paves the way for sustainable medical waste management and flexible load modulation in nuclear power generation.

The detailed energy flow diagrams for the two operating modes are depicted in Figs. 3 and 4, respectively. The novel hybrid system exhibits a heightened magnitude of energy flux. The input of medical waste, quantified at a constant 12.46 MW, serves as the baseline reference at 100 %, enabling an investigation into the efficiency of energy flow through each component. Upon implementation of the upgraded system, 12.46 MW of medical waste energy is introduced into the system. The medical waste undergoes conversion into synthetic gas within the plasma gasification reactor, with a substantial power input (1.62 MW) facilitating plasma torch operation. Consequently, 14.2 MW of energy is transferred to the raw syngas. Through the cooling system, 1.39 MW of thermal energy is channeled to heat steam within the SMR power plant, thereby augmenting power generation. Subsequently, the syngas, boasting 12.81 MW of energy, progresses through the WGS, purification unit, and PSA apparatus to yield hydrogen (H_2) possessing 11.02 MW of energy and 0.03 MW of CO_2 . Evidently, the preeminent energy losses within the entire hybrid system occur predominantly within the WGS and PSA components. In the nocturnal operational mode, a portion of steam is allocated to facilitate heating for the Cu-Cl cycle, resulting in a reduction of power generation to 6.03 MW, albeit with a corresponding 1.90 MW increase in the energy output of hydrogen. With minimal losses incurred, electrical power is transmuted into storable energy in the form of hydrogen.

The present study has also compared the waste-to-energy efficiency of the proposed hybrid system with other projects, including a conventional waste-to-energy plant [34], an integrated plasma gasification fuel cell system [35], plasma gasification hydrogen generation system [36], plasma gasification steam cycle integrated molten carbonate fuel cell system [37], two-stage fluidized bed-plasma gasification integrated with SOFC and chemical looping combustion [38], as illustrated in Fig. 5. The waste-to-energy efficiency of this hybrid system outperforms other gasification-based waste-to-energy systems reported in the literature. Consequently, the proposed scheme represents an efficient energy conversion technology based on comparative results.

4.3. Exergy analysis

The transformation of energy within a system is not solely confined to its quantitative aspect; rather, it hinges on the quality and availability of energy. In this context, exergy is a pivotal metric for assessing available energy. It unveils losses incurred during energy conversion and presents opportunities for system optimization. Exergy analyses were conducted on the SMR power plant and the proposed hybrid system, as depicted in Table 12. The reference unit's exergy input for the SMR power plant was constant, denoted as 100 %. Due to the integration of the proposed system, an additional exergy input of 12.46 MW from medical waste was introduced. The tabulated data reveals that under daytime operational mode, the overall exergy losses for the system amount to 17.82 MW, while under nighttime operational mode, these losses escalate to 18.23 MW. The system-wide exergy efficiency for the two operational modes is 45.31 % (daytime) and 44.17 % (nighttime). The diminishment in exergy efficiency from daytime to nighttime is minimal, ensuring efficient peak-shaving capabilities for the system. Notably, the overall exergy efficiency is enhanced compared to the reference SMR power plant's exergy efficiency of 34.03 %. As illustrated in Fig. 6, within the hybrid system, the SMR power plant exhibits the highest irreversibility, followed by the PSA subsystem, accounting for 17.9 % of the total exergy losses. Discrepancies in temperature between

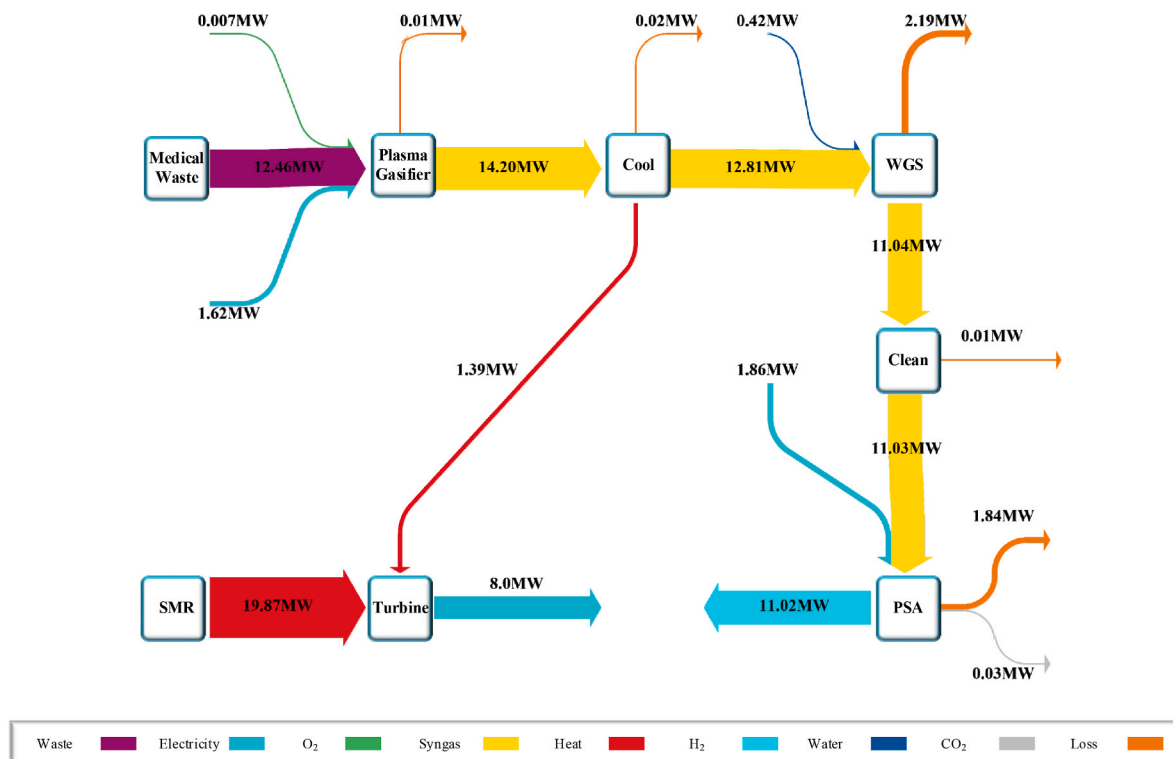


Fig. 3. Energy flow in daytime operation mode.

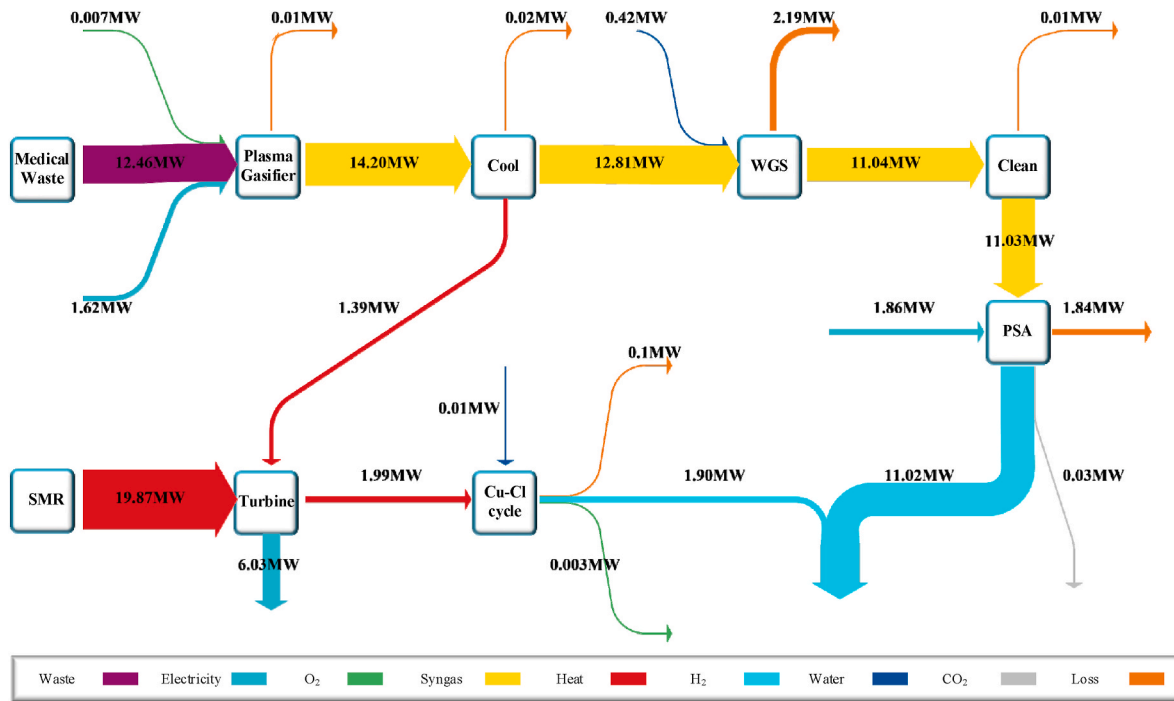


Fig. 4. Energy flow in night operation mode.

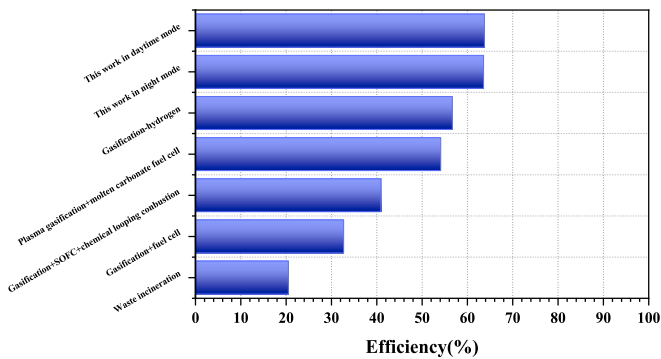


Fig. 5. Comparison of efficiency of various waste-to-energy systems.

Table 12
Exergy performance of the proposed hybrid system.

Item	Reference nuclear plant	Proposed system	
		day	night
	MW	MW	MW
Exergy input from small modular reactor	19.87	19.87	19.87
Exergy input from medical waste	/	12.46	12.46
Exergy input from water	/	0.26	0.34
Total exergy input	19.87	32.70	32.78
Exergy loss	/	1.26	1.26
Plasma gasification subsystem	/	0.14	0.14
SGCs	/	3.19	3.19
PSA subsystem	/	13.23	13.04
Small modular reactor subsystem	/	/	0.60
Cu-Cl cycle subsystem	/	/	0.60
Total exergy loss	13.11	17.82	18.23
Total exergy efficiency	34.03 %	45.31 %	44.17 %

the feed of the plasma gasifier and the high-temperature plasma lead to a significant exergy loss of 1.26 MW.

4.4. Economic analysis

An economic evaluation was carried out to determine the financial viability of the proposed project. As this project is assumed to be a retrofit on an existing SMR power plant, the costs and revenues of the SMR power plant were held constant while separately examining the economics of the medical waste-to-energy conversion and Cu-Cl cycle hydrogen production processes. As shown in Table 13, the project's lifecycle was set at 20 years, with two years allocated for the construction phase and the remaining 18 years for the operational phase. During the 18-year operational phase, the equipment operates for 7920 h per year, with annual operational and maintenance costs amounting to 10 % of the total investment cost at a discount rate of 8 %. Due to the inherent risk of medical waste, its disposal cost is notably high, at 463.86 \$/t. The revenue from H₂ production amounts to 3258.43 \$/t, while the prices for electricity and slag sales are 67.38 \$/MWh and 53.78 \$/t, respectively.

A comprehensive economic evaluation was conducted for the integrated system. Economic assessments were performed individually for the additional components incorporated into the hybrid system, apart from the SMR power plant, which included the plasma gasification sub-system, PSA sub-system, and Cu-Cl cycle sub-system. Investment costs were estimated utilizing the scale factor method, as depicted in Table 14. The formula for the scale factor method is as follows [46].

$$CC_{\text{cur}} = CC_{\text{ref}} \times \left(\frac{S_{\text{cur}}}{S_{\text{ref}}} \right)^f \quad (16)$$

where CC_{cur} and CC_{ref} are capital cost of the current equipment and the reference capital cost, USD; S_{cur} and S_{ref} are the scale parameter of the current equipment and the reference scale parameter; f is the scale factor.

Fig. 7 and Table 15 depict the required investment costs for the hybrid system, which incorporates numerous additional components, such as the plasma gasifier, heat exchangers, Cu-Cl cycle equipment,

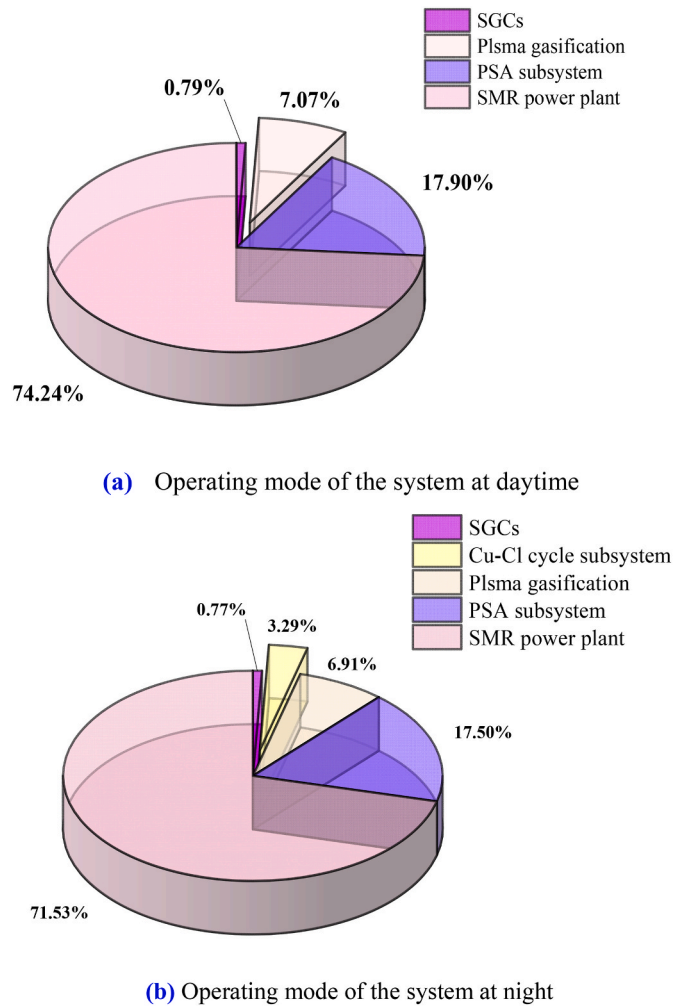


Fig. 6. Exergy loss distribution diagram of individual components.

Table 13
Foundational presumptions and essential data for the economic assessment.

Item	unit	value
Annual operational duration of the project [39]	h	7920
Lifetime of the project	year	2
Economic period [39]	year	18
Price for the disposal of medical waste [40]	\$/t	345.75
Selling price of production H ₂ [41]	\$/t	3258.43
Selling price of vitrified slag [42]	\$/t	53.78
Selling price of grid purchase [43]	\$/MWh	59.44
Maintenance & operating cost [44]	–	10 % of total capital cost
Discount rate [45]	–	8 %

Table 14
Capital cost estimation methods for the components.

Component	Reference capital cost CC_{ref} (k\$)	Reference scale S_{ref}	Scale unit	Scaling factor f	source
Plasma gasifier	78000	39.2	kg/s	0.67	[47]
Cleaning section	159600	20.25	kmol/s	0.7	[48]
Cu–Cl equipment	125000	790	t/day	0.6	[49]
SGC	702.68	13149	m ²	0.68	[50]
HX	702.68	13149	m ²	0.68	
WGS reactor	16790	1377	MW	0.67	[51]
Syngas compressor	4900	20.83	t/h	0.85	

PSA unit, and WGS reactor, totaling an investment of 26,083.75 k\$.

The economic assessment of the new design was conducted using the previously mentioned evaluation method, and the outcomes are presented in Table 16. Medical waste treatment costs are relatively elevated due to the hazardous characteristics of such waste, amounting to an annual expenditure of 7886.42 k\$. Simultaneously, revenue generated from the production of H₂, ash, and other products totals 9124.81 k\$ and 178.52 k\$, respectively. Since the electrical income of the SMR power plant remains unchanged and does not participate in the economic assessment of the project, the addition of new equipment results in increased electricity consumption. Therefore, the annual revenue generated from electricity production amounts to –1640.35 k\$. During the operational phase, annual operational and maintenance expenses amount to 2608.38 k\$, resulting in an annual net income of 13,066.06 k\$. When considering discounted cash flows and excluding the two-year construction period, the investment cost is recouped in just 2.34 years. Over a 20-year operational period, the project's NPV accumulates to 81727.18 k\$. Furthermore, the IRR for this design scheme stands at a robust 41 %. These results underscore the economic viability of the new design.

The economic performance of the two SMRs alone was also calculated. The key economic parameters used in this calculation are referred to in reference [52, 53]. The results of these calculations are presented in Table 17. Comparatively speaking, the performance of the hybrid system is higher than the economic performance of using the two SMRs separately, which confirms the economic feasibility of the new design.

Given the disparate scales of various projects, a direct and intuitive comparison of their respective NPV is not practical. For this reason, the present study employs the DPP to comparatively ascertain the economic benefits of different systems, including hybrid power generation systems with wind power and photovoltaic and fuel cells [54], solar integrated with biomass chemical cycle [55], IGCC combined chemical cycle [56], photovoltaic combined fuel cell [57], plasma gasification for hydrogen [58]. Additionally, the IRR is harnessed as a metric for the economic reliability of these projects. Results delineated in Fig. 8 explicitly reveal that the research provides distinct advantages in terms of both economic viability and reliability.

4.5. Sensitivity analysis

4.5.1. Influence of plasma temperature

The plasma temperature is a critical parameter in the plasma gasification process. Fig. 9 illustrates the impact of plasma temperature on the performance of the novel system. An increase in temperature from 4000 to 4250 causes a slight decrease in the low heating value of the syngas while simultaneously improving the efficiency of the gasification process. However, the medical waste-to-energy efficiency subsequently decreases. Although higher plasma temperature facilitates the molecular decomposition of medical waste, they also increase the auxiliary power consumed by the plasma torch, ultimately reducing the medical waste-to-energy efficiency.

4.5.2. Influence of moisture content in medical waste

Medical waste exhibits a significant moisture content. As shown in Fig. 10, which investigates the effect of medical waste moisture content

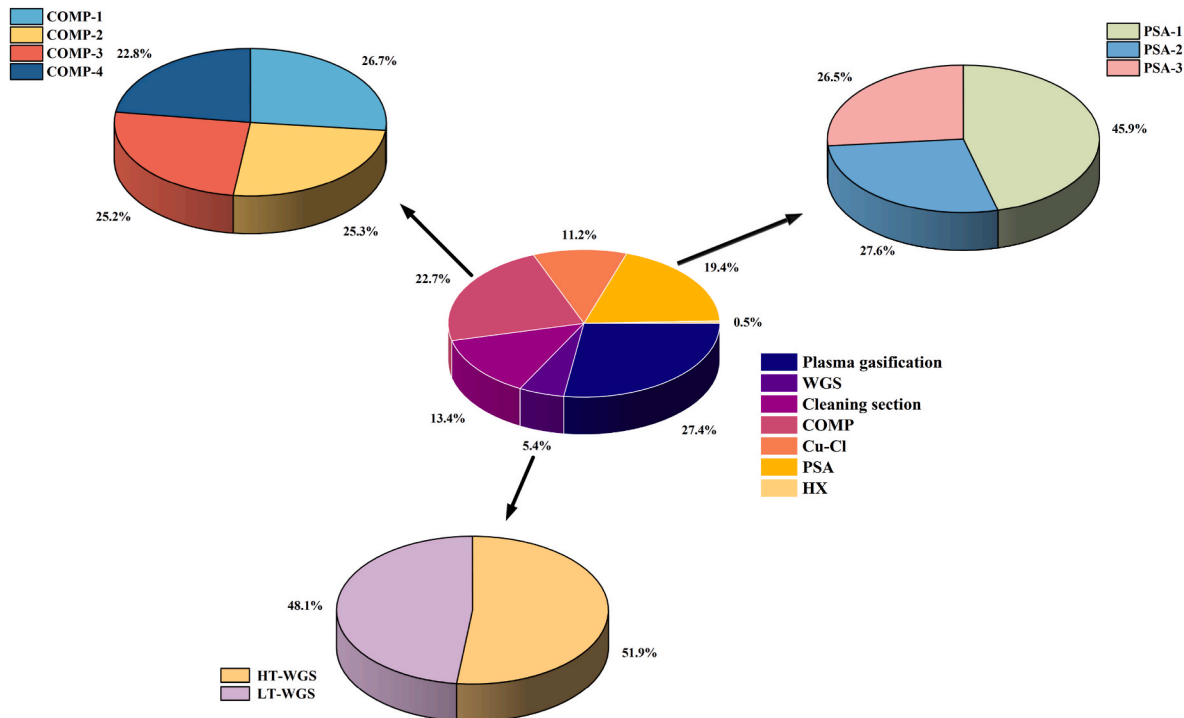


Fig. 7. Diagram of the investment costs of the various components of the proposed project.

Table 15
Investment costs of the components.

Item		Capital cost CC_{cur} (k\$)
Plasma gasification		7138.62
SGCs	SGC-1	42.21
	SGC-2	3.54
WGS reactor	HT-WGS	731.18
	LT-WGS	678.54
Heat exchanger	HX-1	76.31
	HX-2	16.37
Cleaning section		3507.68
Syngas compressor	COMP-1	1582.60
	COMP-2	1498.89
	COMP-3	1494.68
	COMP-4	1347.69
PSA unit	PSA-1	2317.04
	PSA-2	1393.88
	PSA-3	1337.39
Cu-Cl		2917.15
Total investment cost		26083.75

Table 16
Economic analysis results of the proposed system.

Item	unit	value
Annual revenue from medical waste disposal	k\$	7886.42
Annual revenue from production H_2	k\$	9124.81
Annual revenue from slag sale	k\$	178.52
Annual revenue from electricity	k\$	−1640.35
Annual operation and maintenance cost	k\$	2608.38
Net annual income	k\$	13066.06
DPP	year	4.34
NPV	k\$	81727.18
IRR	%	41

on the performance of the proposed new system. The low heating value of the syngas produced by the plasma gasifier decreased with the increase in the moisture content of the medical waste, and the gasification efficiency also decreased. This decrease is primarily attributed to the

Table 17
Economic analysis results of two SMRs.

Item	unit	value
DPP	year	5.53
NPV	k\$	14327.69
IRR	%	25 %

concurrent increase in water content within the syngas. Consequently, the medical waste-to-energy efficiency decreases in both modes. During the design phase of a novel system, it is essential to consider pre-drying raw medical waste as an effective approach to managing high-moisture medical waste.

4.6. Further discussion

Future research will focus on the technical feasibility and scalability of integrating Gen IV reactors with existing hydrogen production processes; perform economic benefit analyses to evaluate the cost-efficiency of the transition from current to Gen IV technologies; investigate the environmental impacts of Gen IV technology to ensure alignment with sustainable ecological practices, to optimize the use of nuclear technology. This effort aims to harness the potential of Gen IV technology for low-carbon, efficient hydrogen production, driving innovation and sustainability in the energy sector.

5. Conclusions

To enhance the peak shaving capability of the SMR and to environmentally manage hazardous solid waste, this study has undertaken the reconfigurable design of a hybrid system building upon the foundation of the SMR power plant. This conceptual hybrid configuration encompasses the SMR power plant, a plasma gasification subsystem, a PSA subsystem, and a Cu-Cl cycle subsystem. Comprehensive technical and economic performance analyses of the hybrid system were carried out. Several key findings emerge from this investigation.

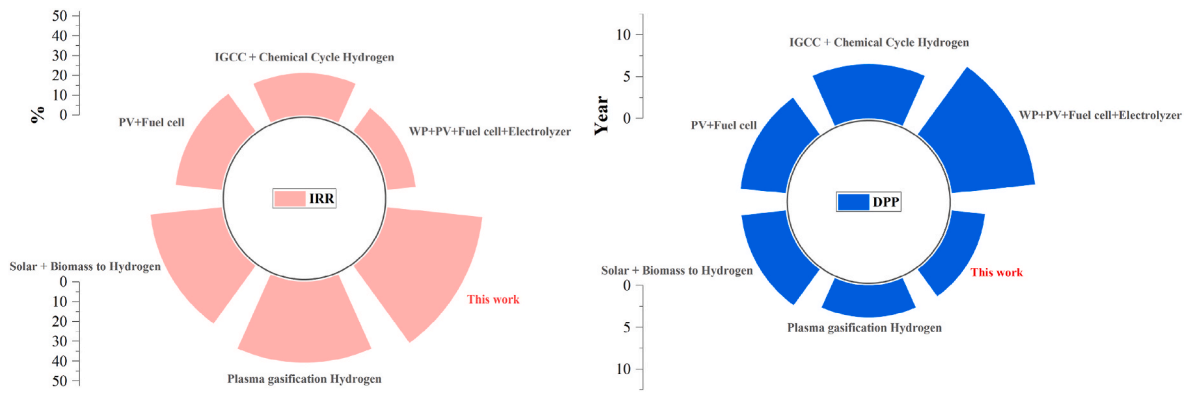
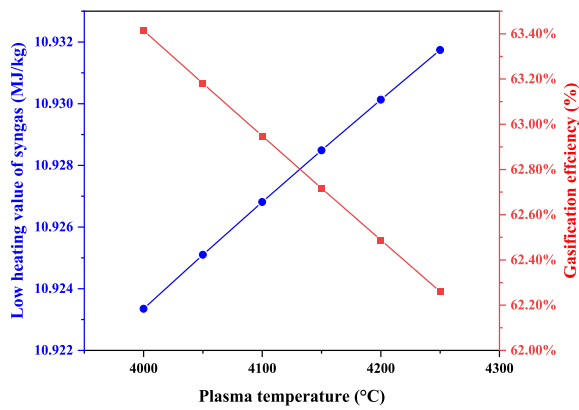
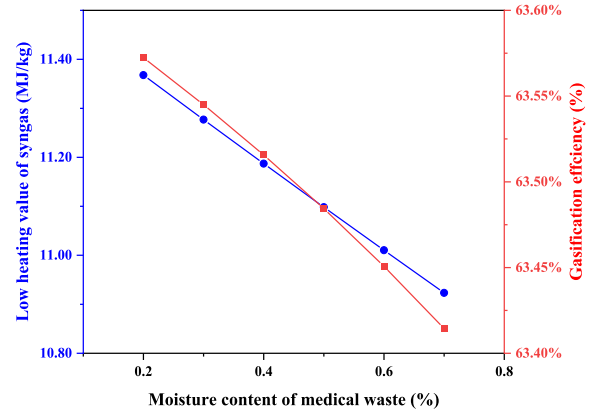


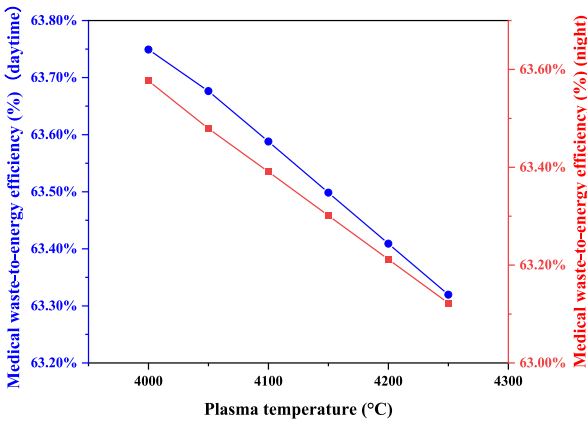
Fig. 8. Comparison of IRR and DPP of various systems.



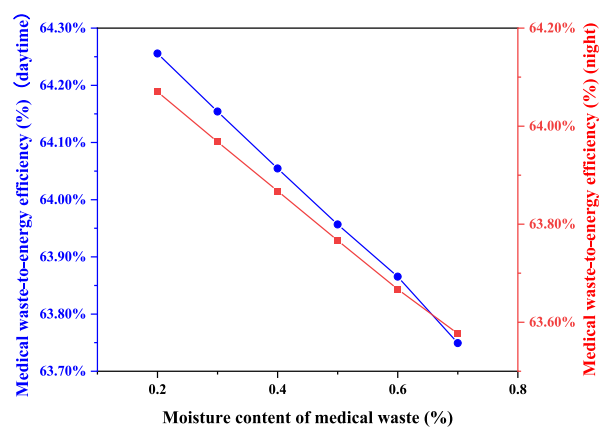
(a) Lower heating value of syngas and plasma gasification efficiency



(a) Lower heating value of syngas and plasma gasification efficiency



(b) Medical waste-to-energy efficiency



(b) Medical waste-to-energy efficiency

Fig. 9. Influence of the plasma temperature on the performance of the proposed hybrid system.

Fig. 10. Influence of the moisture content of medical waste on the performance of the proposed hybrid system.

(1) When the feedstock inputs for both the SMR power plant and the plasma gasifier remain constant, the converted available energy from medical waste in the hybrid system reaches 7.94 MW (daytime) and 7.92 MW (nighttime), with medical waste-to-energy efficiencies of 63.75 % (daytime) and 63.58 % (nighttime) respectively. The system achieves effective peak shaving capabilities of the SMR without significantly compromising efficiency. This thermodynamically substantiates the viability and efficacy of the novel design.

(2) Exergy analysis reveals that through both operational modes, the exergy efficiencies stand at 45.31 % (daytime) and 44.17 % (nighttime). The SMR power plant exhibits the highest irreversibility, followed by the PSA subsystem. The plasma gasification subsystem ranks third in irreversibility, and collectively, these three subsystems contribute significantly to the predominant portion of exergy losses within the system.

(3) The novel design demonstrates robust cost-effectiveness, as evidenced by its financial performance. With a total investment of

26,083.75 k\$, the project exhibits an NPV of 81,727.18 k\$ over its 20-year lifecycle. Moreover, the proposed innovative design boasts a DPP of merely 4.34 years and an IRR of 41 %. These findings collectively underscore the economic prowess of the suggested hybrid system, highlighting its feasibility within the realm of financial considerations.

CRediT authorship contribution statement

Haoran Wu: Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Heng Chen:** Writing – review & editing, Conceptualization. **Lanxin Fan:** Software, Resources. **Peiyuan Pan:** Writing – review & editing, Validation. **Gang Xu:** Software, Methodology. **Lining Wu:** Validation, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A

Table A1
Chemical reactions take place during the gasification of waste materials [59].

$C + 0.5O_2 \rightarrow CO$	(A1)
$CO + 0.5O_2 \rightarrow CO_2$	(A2)
$C + O_2 \rightarrow CO_2$	(A3)
$H_2 + 0.5O_2 \rightarrow H_2O$	(A4)
$C + H_2O \rightarrow CO + H_2$	(A5)
$CO + H_2O \rightarrow CO_2 + H_2$	(A6)
$CH_4 + H_2O \rightarrow CO + 3H_2$	(A7)
$C + 2H_2 \rightarrow CH_4$	(A8)
$CO + 3H_2 \rightarrow CH_4 + H_2O$	(A9)
$C + CO_2 \rightarrow 2CO$	(A10)
$pC_xH_y \rightarrow qC_nH_m + rH_2$	(A11)
$C_nH_m \rightarrow nC + 0.5mH_2$	(A12)

Table A2
Design parameters of HTR-10 [60].

Item	unit	value
Heat input	MW	10
Main feed water temperature	°C	104
Main steam pressure	MPa	4
Main steam temperature	°C	440
Steam mass flow	Kg/s	3.47





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